

AKSHAYA PATRA

(ak-shy-ya puh-tra) — our Cornucopia

Endless Safe Food Discovery

AI-Powered Dietary-Aware
Food Discovery System

Ushasree Jakilinki

SW Engineer → Aspiring Data Scientist
UofM Master's in Applied Data Science (2025)

TRANSFORMING FOOD ANXIETY

537M

People with diabetes worldwide

33M

Americans with food allergies

3M

Americans with celiac disease

About Me



25+ Years

SW Engineering
Automotive & Mobile



BSc CSE

Computer Science
& Engineering



AI/DS Pivot

Aspiring
Data Scientist



UofM MADS

Master's Applied
Data Science 2025

Why Food Discovery AI?

Firsthand experience navigating food allergies, diabetes, and blood pressure dietary needs — shopping across Amazon, Walmart, Kroger, Instacart weekly and never once receiving a useful dietary-aware recommendation. Buying patterns tell the story. Nobody is connecting the dots.

This Is Just My Immediate Family.

Me

Family history of diabetes · PCOS ·
Anaphylaxis to dairy

My Mom

High blood pressure — low sodium diet
essential

My Brother

Anaphylaxis to shellfish

My Nieces

Allergic to all nuts

My Brother-in-law

Allergic to eggs

Most Family Elders

Both high blood pressure AND diabetes
— two dietary constraints in one
grocery cart

Food restrictions are not a niche problem. They are in every family — including mine. Every one of these people shops for groceries every week. With no intelligent help.

Why Food?

Groceries are the only major consumer purchase that does all of this — every single week.

Weekly. For life.

No other major purchase recurs this often. Groceries are the only buying decision you make every single week, for your entire life, without exception.

Direct health consequences.

A bad Netflix recommendation wastes 2 hours. A bad food recommendation can trigger anaphylaxis, spike blood sugar, or damage intestinal lining. The stakes are unique.

Invisible ingredients.

A product's name tells you nothing about what is inside. Dairy hides under 12+ names. Gluten hides in soy sauce. The label is the only truth — and nobody is reading it for you.

400,000+ products. Structured data. Unused.

The FDA mandates standardised nutrition facts and ingredient lists on every product. USDA tracks 400,000+ branded foods. This is the most structured food dataset in the world. Nobody is mining it for you.

Avoidance patterns are invisible.

Netflix knows what you watched. Spotify knows your mood. Walmart knows you buy coconut milk every week and never buy regular milk — and still recommends cheese. What you avoid tells the story.

Netflix knows what you watched 3 years ago. Spotify knows your 6am mood. Walmart knows you buy coconut milk every week — and still recommends cheese. That gap is Akshaya Patra.

' *Annam Parabrahma Swaroopam* '

Food is the form of the Supreme Being — Taittiriya Upanishad

Universal

Food is how every culture on earth celebrates birth, marks death, seals friendship, and heals the sick. It is the most human act there is.

Science

Food is medicine. What you eat every week directly determines your risk of diabetes, heart disease, cancer, and inflammation. Diet is the single most modifiable health factor.

2026

Yet in 2026, the most important weekly purchase you make — the one with the most data, the most health consequence, and the most personal meaning — gets the least intelligent help of any retail category.

The Never-Ending Circle of Buying Food

Every. Single. Week. For every person with a dietary restriction.

1

Need food

Family needs to eat.
Budget, preferences,
restrictions all in play.

2

Search & check labels

Scan every product.
Google every ingredient.
Is casein dairy?

3

Find something safe

Maybe. Often the same
5 products. Rarely
something new.

4

Cook at home

Too scared to trust
packaged food. Make
everything yourself.

5

Repeat next week

Formulas change.
Products disappear.
Starts all over again.

The Scale of the Problem

The haystack is massive. The needle is findable. Nobody is looking.

500K–1M+

Distinct grocery SKUs & variants

Total US grocery universe

20K–50K

**Grocery items in a typical Walmart
store**

Store-level daily assortment

100K+

SKUs including online & regional

Broader Walmart assortment exposure

So why can't anyone find the right food?

With 500,000 to 1,000,000+ grocery SKUs in the US market and 100,000+ available through Walmart alone, finding the right product for someone with dietary restrictions, sensitivities, or food-based conditions should be entirely doable. The data exists. The products exist. What is missing is a system that connects a person's dietary needs to the right product discovery. That is exactly what Akshaya Patra builds.

The Problem



1. Allergens

Life-threatening reactions hidden in everyday food. Milk alone appears under 12+ names on ingredient lists — casein, whey, ghee and more. Soy, wheat & milk appear in 73-75% of processed foods. 33 million Americans affected. See slide 12.



2. Food-Based Diseases

Diabetes, celiac, high blood pressure, PCOS — all managed primarily through what you eat every week. Hundreds of millions affected globally. Yet no grocery platform accounts for this.

Diabetes — The Fastest Growing Global Public Health Concern

537M

People with diabetes in 2021

10.5% of world population

783M

Projected by 2045

46% increase in 24 years

~50%

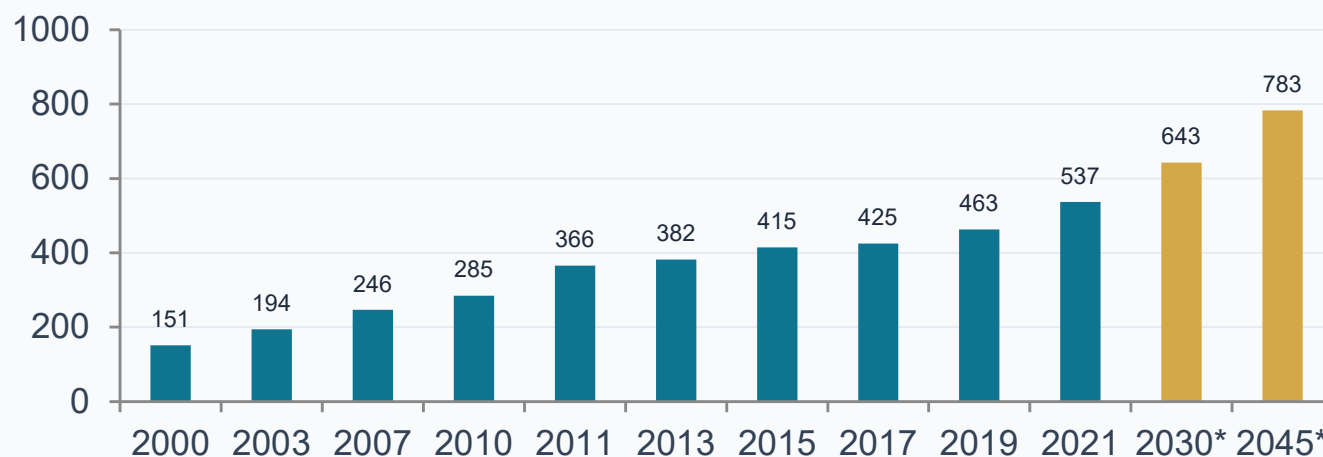
Are undiagnosed

240M people unaware they have it

\$966B

Global healthcare cost in 2021

Rising to \$1,054B+ by 2045



Key findings

- Top 10 countries: India, China, USA lead
- Africa: highest undiagnosed rate (53.6%)
- Diet & exercise reduce T2D risk by 31-46%
- Food is the #1 modifiable risk factor

Source: Hossain MJ, Al-Mamun M, Islam MR. Health Sci Rep. 2024;7(3):e2004. DOI: 10.1002/hsr2.2004. PMC10958528 | * = IDF Atlas 10th ed. projections | Note: Latest published global prevalence data is from IDF Atlas 2021 — 2026 figures not yet published.

How Walmart Recommends — And Why It Fails

Walmart's XP Framework

Item Cross-Pollination

- Treats all products equally — food, electronics, clothing, toys
- Recommends based on co-purchase patterns across all categories
- Customers who bought X also bought Y — regardless of what X or Y actually is

X

Why Food Is Different

- X A bad TV recommendation is annoying. A bad food recommendation can hospitalise you.
- X What you never buy tells the story — avoidance patterns are invisible to XP
- X Ingredient complexity is massive — product names hide what is actually inside

Real Example:

A dairy-allergic shopper buys coconut milk every week and never buys regular milk. Walmart's XP engine sees a dairy-aisle purchase and recommends cheese. Our system sees the avoidance pattern and recommends Talenti Dairy-Free at 50% off.

Food Recommendations are Not Product Recommendations

Why a single recommendation engine cannot serve both.

	Non-Food (TV / Clothing / Toys)	Food (Groceries)
Wrong recommendation risk	Annoying. Easy to ignore.	Life-threatening. Anaphylaxis.
Key signal	What you bought	What you bought AND avoided
Ingredient complexity	None	500K+ SKUs, hidden aliases, label gaps
Purchase cycle	Months to years	Every week — every mistake hurts
Personalisation depth	Style / price preference	Medical necessity + preference
Data needed	Purchase history only	Purchase history + nutritional DB + allergen map

Akshaya Patra builds a food-first recommendation engine — because food is not just another product.

Did You Know These Are Dairy-Free?

Most dairy-allergic shoppers don't. Safe products are hiding in plain sight.

DAIRY-FREE

Oreo

No milk. No butter. No dairy of any kind.

America's favourite cookie — safe for most dairy-allergic shoppers.

DAIRY-FREE

Biscoff

100% plant-based. Zero dairy ingredients.

The airline biscuit loved worldwide — hidden gem for dairy-free diets.

DAIRY-FREE

**Lorna Doone
Shortbread**

Despite the buttery taste — no dairy.

Tastes like it shouldn't be safe. The label says otherwise.

The scale makes human discovery impossible.

USDA FoodData Central tracks 400,000+ branded products — each with full ingredient lists, nutrition facts, and label claims. No allergy sufferer can review them all. But AI can. Akshaya Patra surfaces the safe discoveries that are hiding in plain sight.

Safe Discoveries Hide Across All Conditions

My mom has high blood pressure. Finding low-sodium snacks she actually enjoys took months of label reading.

High Blood Pressure

What the condition requires:

- Sodium under 1,500–2,300mg per day (AHA guideline)
- Avoid regular chips, canned soups, processed snacks
- Most snack labels are not shopper-friendly for sodium tracking
- Discovery of low-sodium alternatives is entirely manual

LOW SODIUM

Ruffles Reduced Sodium

Same great taste. Significantly less sodium. A safe snack option hiding in plain sight.

*My mom found this after months of reading labels in the chips aisle.
No app. No recommendation. Just luck.*

Akshaya Patra would have surfaced this on week one.

Allergies. Blood pressure. Diabetes. PCOS. Every condition has safe options hiding in a 400,000+ product database. Nobody is connecting the dots.

Market Opportunity

33M+

Americans with food allergies, celiac & Alpha-gal

33M allergies · 3M celiac · 450K+ Alpha-gal

\$19B

Buying power of allergy sufferers alone

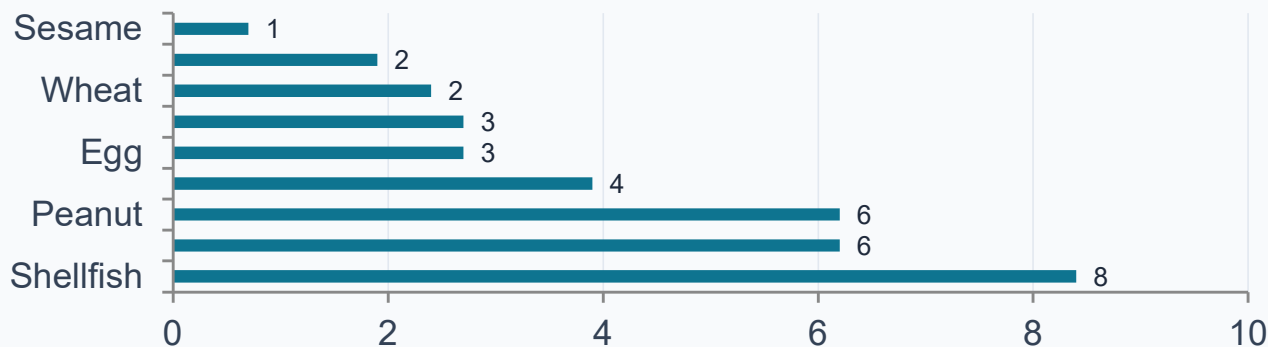
FARE — food allergies only; celiac & Alpha-gal add billions more

\$25B

Annual US economic loss from food allergies

Celiac costs an additional \$17B+ annually

Most Common Allergens (US, millions affected)



Source: FARE (Food Allergy Research & Education). Facts and Statistics. foodallergy.org/resources/facts-and-statistics

Extensible to:

- Diabetes
- Celiac disease
- Alpha-gal syndrome
- Vegan / Keto / Low-carb
- Pancreatitis / IBD

The Market — Food Visibility Matters for All of Them

Every condition below is managed through weekly food choices. Every person affected needs food visibility. Nobody is providing it.

33M

Americans affected

Food Allergies

Life-threatening reactions. Hidden in 73-75% of processed foods.

3M

Americans affected

Celiac Disease

Autoimmune. Only treatment is 100% gluten-free diet. 50-83% undiagnosed.

119.9M

Americans affected

High Blood Pressure

Nearly 1 in 2 adults. Low-sodium diet is primary lifestyle intervention.

1 in 7

Women of reproductive age

PCOS

75% undiagnosed. Diet is a primary management tool. \$8B annual US healthcare cost.

537M

People worldwide

Diabetes

Projected 783M by 2045. High protein, high fibre, low carb diet is key.

Combined market: hundreds of millions of people making dietary decisions every week — with no intelligent food discovery to help them.

Sources: FARE (foodallergy.org) | Beyond Celiac (beyondceliac.org) | CDC (cdc.gov/high-blood-pressure) | Am J Obstet Gynecol 2023 ([doi:10.1016/j.ajog.2023.04.010](https://doi.org/10.1016/j.ajog.2023.04.010)) | Hossain et al. Health Sci Rep 2024 (PMID:38524769)

Why Labels Aren't Enough

Milk alone appears under 12+ names. Our system maps all aliases to a single allergen tag.

Casein

Caseinate

Lactalbumin

Lactoglobulin

Whey

Whey powder

Lactose

Milk solids

Nonfat milk solids

Ghee

Cream

Lactulose

All 12 map to → MILK. Our labelling engine handles multi-alias resolution automatically. The same logic applies to all top 9 allergens — and can extend to vegan, gluten-free, keto, low-sodium, and medical dietary codes.

Real Example

Spring Roll Wrappers

Ingredient: SODIUM CASEINATE (MILK)

*Nobody expects milk in spring roll wrappers.
Learned the hard way.*

The Data Is Already There — FDA Mandated It

Every food label in the US uses FDA-standardised, legally defined nutrient claim words. This is the most structured, machine-readable dataset in the entire food industry. Nobody is mining it for personalised recommendations.

Absence claims

Free, Zero, No, Without

Reduction claims

Low, Reduced, Less, Fewer, Light/Lite, Little

Positive claims

High, Rich In, Excellent Source Of, Good Source, Contains, Provides

Addition claims

More, Fortified, Enriched, Added

Lifestyle labels

Healthy, Vegan, Keto, Organic, Whole Grain, Non-GMO, Plant-Based

Diet patterns

Low-carb, High-protein, Mediterranean, DASH, Low-FODMAP, Diabetic-friendly

Akshaya Patra reads the label so you do not have to — and matches it to your dietary needs automatically.

The Solution

A 3-step AI pipeline that turns grocery purchase history into personalised, allergy-safe recommendations.

01

Predict Sensitivities

Multi-label classifier
infers allergen risk
from purchase history

Prediction AI

02

Label Products

Rule-based engine tags
all SKUs with binary
allergen flags

Rules + DB

03

Recommend Safely

Ranking layer combines
steps 1 & 2 with sales,
novelty & discounts

Recommendation AI

Steps 1 & 2 are independent and can run in parallel — Step 1 is the priority as it requires iterative data mapping.

Step 1 — Predict Allergen Sensitivities

INPUTS

- 6 months of purchase history
- USDA / Store nutritional DB
- User account preferences (optional)

Multi-Label Classifier

Prediction AI

OUTPUT

Probability score per allergen
per user (0.0 → 1.0)

*Exact Match Ratio
Hamming Loss · Precision*

Sample Output — Sensitivity Scores (0 = none, 1 = high)

User	Milk	Tree Nuts	Wheat	Soybeans
User-1	0.96 ▲	0.02	0.03	0.03
User-2	0.01	0.99 ▲	0.03	0.01
User-3	0.02	0.10	0.03	0.87 ▲
User-4	0.03	0.10	0.99 ▲	0.01

Step 2 — Product Allergen Labelling

A rule-based system that scans every product's ingredient list and assigns binary Y/N allergen tags. Not AI — pure data engineering. Critical foundation for Step 3.

Key Challenge

One allergen = many names.
Milk alone appears as: casein, whey, lactalbumin, lactoglobulin, caseinate...

Sample Output — Product Allergen Tags

Product	Milk	Eggs	Wheat	Soy	Tree Nuts
Oreo	N	N	Y	Y	N
Biscoff	N	N	Y	Y	N
Shortbread	Y	N	Y	Y	N
Almond Croissant	Y	Y	Y	Y	Y
Oranges	N	N	N	N	N
Talenti Dairy-Free Ice Cream	N	Y	N	Y	Y

Input: USDA Nutritional DB or store-level DB | Data source: Walmart API, GroceryDB, Instacart

Step 3 — Smart Recommendations in Action

The ranking layer combines Steps 1 & 2 with sales signals to surface safe, relevant, timely products.

Current Promotions

Almond Croissant **10% off**

Oranges **25% off**

Talenti Dairy-Free **50% off**

+ Step 1 &
2
—
Outputs

Personalised Recommendations

User-1 High Milk → Talenti Dairy-Free + Oranges

User-2 High Tree Nuts + Peanuts → Oranges only

User-3 High Soybeans → Oranges only

User-4 High Wheat → Talenti Dairy-Free + Oranges

User-5 None detected → Almond Croissant (50% off!)

Data Sources & Research Foundation

Transaction Data

- Instacart Online Grocery Dataset (Kaggle)
- Dunnhumby Complete Journey Dataset
- Walmart+ / Kroger / Amazon purchase history

Nutritional DBs

- USDA FoodData Central
- GroceryDB (Barabasi Lab, Harvard)

Research Papers

- Grocery to General Merchandise: XP Recommender (arXiv 2509.02890)
- Hossain MJ et al. Diabetes mellitus. Health Sci Rep. 2024 (PMC10958528)

Allergy & Health Orgs

- FARE (Food Allergy Research & Education) — foodallergy.org
- Beyond Celiac — beyondceliac.org

Research Questions



M1

Can we identify dietary-restricted users from purchase patterns alone?

Multi-label classification on grocery transaction history. No explicit user-declared preferences needed.



M2

How accurately can we tag products with allergen flags at scale?

Rule-based NLP pipeline over ingredient lists. Precision / recall against USDA ground truth.



Capstone

Does a constraint-aware recommendation engine improve safe food discovery?

End-to-end system: Hit Rate, NDCG, Revenue-per-user, and user-reported allergy safety satisfaction.

Success Metrics

Step	Metric	Why It Matters
Step 1	Exact Match Ratio	All allergens for a user must be predicted correctly
Step 1	Hamming Loss	Penalises each individual allergen mislabelling
Step 1	Precision / Recall	False negatives are dangerous — must minimise missed allergens
Step 2	Label Accuracy vs USDA	Ground truth validation of allergen tags
Step 3	Hit Rate @ K	Did we surface at least one product the user would buy?
Step 3	Revenue per User	Business impact of better discovery
Step 3	Avg Order Value	Higher when users find what they want
Step 3	Latency	Real-time ranking must be < 200ms

Technical Approach

Component	Approach	Data
Step 1 — Sensitivity Prediction	Multi-label classifier (e.g. Binary Relevance, Label Powerset, or Transformer-based)	Instacart / Dunnhumby purchase history + USDA ingredient DB
Step 2 — Product Labelling	Rule-based NLP + regex over ingredient lists; allergen synonym dictionary	USDA FoodData Central, GroceryDB, Walmart Item API
Step 3 — Recommendation	Constraint-aware ranking layer: filters unsafe products, boosts sale/new items	Outputs of Steps 1 & 2 + real-time sales/promo feed
Evaluation	Exact Match Ratio, Hamming Loss, Precision/Recall, Hit Rate@K, NDCG	Held-out user purchase data, USDA ground truth labels

Skills Demonstrated

✓ Multi-label classification

✓ NLP & text parsing

✓ Recommendation systems

✓ Data engineering

✓ Product thinking

✓ Dietary data ethics

Food IS the Intervention

For diabetes, what you eat every week directly impacts your health outcomes. Yet grocery recommendations ignore this entirely.



High Protein

Slows glucose absorption. Promotes satiety.
Reduces overeating and blood sugar spikes after meals.

e.g. Eggs, lentils, chicken, Greek yogurt, tofu, fish



High Fibre

Slows carbohydrate digestion. Stabilises blood sugar. Feeds gut microbiome linked to insulin sensitivity.

e.g. Vegetables, legumes, whole grains, berries, nuts, seeds



Low Carb

Directly reduces glucose load. The single most impactful dietary lever for blood sugar management.

e.g. Avoid refined sugar, white bread, white rice, sugary drinks

Personal example — this is the dietary approach I use. Food discovery that understands these constraints is not a luxury. It is a weekly necessity.

Extensibility — Conditions & FDA Label Intelligence

The same 3-step architecture extends to any dietary need, medical condition, or FDA-regulated label claim.

Celiac / Gluten-Free

Gluten appears in wheat, barley, rye, malt — and in medications

Diabetes

Glycemic index, carb load, sugar substitutes — monk fruit cookie alerts

Alpha-gal Syndrome

450K possibly undiagnosed — avoids all mammalian meat products

Vegan / Vegetarian

Hidden animal derivatives in broths, fats, and additives

Low-Sodium / Heart

Sodium content across 100K+ SKUs — personalised threshold alerts

Pancreatitis / IBD

Low-fat, low-fiber requirements shift by symptom severity

G6PD deficiency / Fava beans (favism)

Inherited genetic disorder — avoids fava beans and specific drugs

Histamine intolerance

Aged & fermented goods

Kidney disease(CKD)

Potassium · phosphorus · sodium

One engine, every condition — all detectable from FDA-mandated ingredient & nutrition data (e.g. the required “CONTAINS PHENYLALANINE” label).

The Platform Extends Further

Beyond conditions and conditions — the architecture reads any structured label data.

FDA-Standardised Label Claims — Already on Every Product

Every food product in the US must use legally defined, standardised claim words. This is the most structured, machine-readable dataset in the food industry — and it is already on every label.

Absence

Free · Zero · No · Without

Reduction

Low · Reduced · Less ·
Fewer · Light / Lite · Little

Positive

High · Rich In · Excellent
Source Of · Good Source ·
Contains · Provides

Addition

More · Fortified · Enriched ·
Added

Lifestyle

Healthy · Organic · Whole
Grain · Vegan · Keto · Non-
GMO · Plant-Based · DASH ·
Low-FODMAP

Pet Food — Same Problem, Entirely New Market

90M+

US dogs

\$50B+

Pet food market

12+

Names for chicken

Chicken alone hides as: poultry by-product · animal digest · chicken meal · chicken fat · dried poultry · mechanically separated chicken. AAFCO standardised labels already exist — the same 3-step architecture applies directly.

Vision & Next Steps

Phase 1 — Now

- Build & validate multi-label classifier (M1)
- Allergen tagging pipeline for top 9 allergens
- Proof-of-concept recommendation on Instacart data

Phase 2 — Capstone

- End-to-end constraint-aware recommendation system
- Extend to celiac, diabetes, vegan labels
- Partner outreach: Beyond Celiac, FARE, foodallergy.org

Phase 3 — Vision

- Real-time checkout integration with retail partners
- Proactive sale alerts for safe products
- Ph.D. research: clinical validation with health outcomes data

"In the Mahabharata, the Akshaya Patra was a divine vessel that provided endless nourishment to those in need. We channel that same principle — boundless abundance for people with food restrictions."

Let's Connect

AKSHAYA PATRA · Endless Safe Food Discovery

Ushasree Jakilinki

linkedin.com/in/ushasreejakilinki

github.com/auj012/PublicReporforSharing

Aspiring Data Scientist · UofM MADS 2025 · Passionate about AI for preventive health